

EDMONTON INTL, AB, Canada

WMO: 711230

Lat: 53.3067N

Lon: 113.6058W

Elev: 2373

StdP: 13.48

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -26.2	(c) -20.1	(d) -33.6	(e) 0.9	(f) -26.0	(g) -27.5	(h) 1.3	(i) -20.0	(j) 25.3	(k) 25.4	(l) 22.4	(m) 18.2	(n) 4.6	(o) 190	(p) 0.685

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.9	82.0	64.4	79.0	63.2	76.1	61.5	67.8	78.2	65.5	75.7	63.5	73.0	8.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.0	97.6	73.3	61.6	89.4	70.6	59.6	83.2	68.6	33.6	78.4	31.7	75.6	30.1	73.0	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	20.0	17.5	-36.1	88.3	7.3	3.5	-41.3	90.9	-45.6	92.9	-49.7	94.9	-55.0	97.5
			-36.3	72.1	7.2	2.9	-41.5	74.1	-45.7	75.8	-49.8	77.4	-55.0	79.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.4	10.3	12.9	22.4	38.1	50.2	57.3	61.4	59.1	50.8	38.5	22.8	12.0
	DBStd	21.56	16.05	13.95	14.21	9.54	8.08	5.42	5.10	5.66	7.64	8.29	12.59	14.52
	HDD50	6037	1232	1038	856	365	97	7	1	4	80	364	816	1177
	HDD65	10465	1697	1458	1320	806	461	237	135	194	428	821	1266	1642
	CDD50	1089	0	0	0	9	103	226	352	287	104	8	0	0
	CDD65	43	0	0	0	0	2	6	22	12	1	0	0	0
	CDH74	1096	0	0	0	4	105	150	396	314	123	4	0	0
	CDH80	196	0	0	0	0	17	21	75	58	25	0	0	0
	WSAvg	7.8	7.4	7.5	8.2	9.3	9.5	8.3	6.9	6.4	7.4	8.2	7.5	7.5
	PrecAvg	19.4	1.0	0.6	0.7	1.0	1.9	3.3	3.9	2.8	1.9	0.8	0.7	0.7
(15) Precipitation	PrecMax	25.6	2.9	1.6	1.8	2.4	5.4	6.7	8.2	7.8	5.7	2.7	2.4	1.8
	PrecMin	14.1	0.2	0.0	0.1	0.1	0.4	0.5	0.7	0.3	0.1	0.0	0.0	0.1
	PrecStd	3.0	0.7	0.4	0.4	0.6	1.0	1.7	1.5	1.6	1.2	0.6	0.5	0.4
	PrecStd	3.0	0.7	0.4	0.4	0.6	1.0	1.7	1.5	1.6	1.2	0.6	0.5	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	43.1	43.4	54.4	72.8	82.4	82.8	86.5	85.7	83.5	71.4	54.6	45.1	
	0.4% MCWB	36.8	36.5	44.0	52.4	57.8	63.4	68.4	66.8	62.4	54.1	44.6	37.6	
	2% DB	39.2	38.7	47.1	66.0	76.6	78.1	81.7	80.9	77.4	63.1	46.7	39.8	
	2% MCWB	35.0	33.9	39.0	48.7	55.6	60.7	67.0	65.1	59.5	49.9	39.5	34.6	
	5% DB	35.8	35.6	42.3	60.2	72.4	74.6	78.4	77.4	72.4	57.4	41.9	36.3	
	5% MCWB	32.6	31.6	36.5	45.8	53.9	59.4	65.7	63.6	57.5	46.4	36.4	32.2	
	10% DB	31.8	32.3	38.9	55.0	67.8	71.1	75.2	74.0	67.2	52.9	38.1	32.2	
	10% MCWB	29.0	29.2	34.3	43.0	51.6	57.6	63.4	62.3	54.9	44.0	34.1	29.2	
	0.4% WB	37.4	37.2	44.5	53.0	59.8	66.6	71.9	69.5	63.8	55.1	45.4	38.4	
	0.4% MCDB	41.7	42.4	52.9	70.2	76.6	76.7	80.7	80.2	78.6	69.7	53.6	44.2	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	35.2	34.4	39.6	49.7	57.2	63.5	68.9	67.0	60.5	50.8	40.4	35.0	
	2% MCDB	38.7	38.2	45.9	64.5	72.4	73.9	79.0	77.7	74.9	61.0	45.7	39.2	
	5% WB	32.7	31.9	36.8	46.7	55.0	61.2	66.6	65.0	58.2	47.5	37.0	32.3	
	5% MCDB	35.8	35.2	41.8	58.7	69.8	71.2	76.5	75.4	71.2	55.8	41.1	36.1	
	10% WB	28.9	29.1	34.7	43.8	52.7	59.1	64.4	63.0	55.6	44.9	34.4	29.1	
	10% MCDB	31.6	32.4	38.5	53.5	65.9	68.3	73.5	72.5	65.6	51.7	37.8	32.2	
(35) Mean Daily Temperature Range	MDBR	18.3	20.3	18.9	22.0	24.8	21.8	22.9	24.5	25.2	22.0	17.4	18.1	
	5% DB	20.4	22.6	21.8	33.0	32.8	29.1	29.2	31.4	35.4	31.0	21.8	20.0	
	5% WB	17.1	18.7	15.4	18.6	16.3	15.4	17.0	17.8	19.4	19.2	15.6	16.1	
	MCDBR	19.8	21.5	20.6	31.0	29.6	24.7	27.0	28.6	33.0	28.5	20.8	19.7	
	MCWBR	16.7	18.0	14.9	18.1	15.2	14.5	16.8	17.4	18.4	18.3	15.4	16.4	
	MCWBR	16.7	18.0	14.9	18.1	15.2	14.5	16.8	17.4	18.4	18.3	15.4	16.4	
(40) Clear-Sky Solar Irradiance	taub	0.257	0.265	0.308	0.326	0.342	0.350	0.353	0.361	0.307	0.282	0.254	0.241	
	taud	2.315	2.373	2.337	2.390	2.393	2.388	2.421	2.354	2.497	2.545	2.461	2.318	
	Ebn at Noon	234	272	279	288	288	285	282	273	279	265	239	219	
	Edh at Noon	18	23	31	33	35	36	34	34	26	20	16	15	
(44) All-Sky Solar Radiation	RadAvg	317	647	1128	1484	1814	1912	1967	1618	1119	649	316	233	
	RadStd	33	51	84	141	128	124	99	111	116	75	35	22	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (73 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page